```c
    for(int i = 0; i < nb; i++) {
        printf("Block %d: ", i + 1);
        scanf("%d", &block[i]);
    }

    // Input process sizes
    printf("\nEnter size of each process:\n");
    for(int i = 0; i < np; i++) {
        printf("Process %d: ", i + 1);
        scanf("%d", &process[i]);
        allocation[i] = -1; // Initially not allocated
    }

    // First Fit Allocation
    for(int i = 0; i < np; i++) {
        for(int j = 0; j < nb; j++) {
            if(block[j] >= process[i]) {
                allocation[i] = j;
                block[j] -= process[i]; // reduce available size
                break;
            }
        }
    }

    // Output
    printf("\nProcess No.\tProcess Size\tBlock No.\n");
    for(int i = 0; i < np; i++) {
        printf("%d\t\t%d\t\t", i + 1, process[i]);
        if(allocation[i] != -1)
            printf("%d\n", allocation[i] + 1);
        else
            printf("Not Allocated\n");
    }

    return 0;
}
```

```
        printf("Process %d: ", i + 1);
        scanf("%d", &process[i]);
        allocation[i] = -1; // Not allocated initially
    }

    // Best Fit Allocation
    for(int i = 0; i < np; i++) {
        int bestIndex = -1;

        for(int j = 0; j < nb; j++) {
            if(block[j] >= process[i]) {
                if(bestIndex == -1 || block[j] < block[bestIndex]) {
                    bestIndex = j;
                }
            }
        }

        if(bestIndex != -1) {
            allocation[i] = bestIndex;
            block[bestIndex] -= process[i]; // reduce block size
        }
    }

    // Output
    printf("\nProcess No.\tProcess Size\tBlock No.\n");
    for(int i = 0; i < np; i++) {
        printf("%d\t\t%d\t\t", i + 1, process[i]);

        if(allocation[i] != -1)
            printf("%d\n", allocation[i] + 1);
        else
            printf("Not Allocated\n");
    }

    return 0;
}
:wq
```

```
nithyasri@nithyasri-QEMU-Virtual-Machine:~/shell/pc/pc/pc/pc$ ls
a.out   bestfit.c  fcf.c   first.sh    firtfit.c  ntihya.sh  roundro.c
backup  bestfit.s  fcfs.c  firstfit.c  nithya.sh  prod.c     script.awk
nithyasri@nithyasri-QEMU-Virtual-Machine:~/shell/pc/pc/pc/pc$ vi bestfit.c
nithyasri@nithyasri-QEMU-Virtual-Machine:~/shell/pc/pc/pc/pc$ ./a.out
Enter number of memory blocks: 7
Enter number of processes: 7

Enter size of each block:
Block 1: 19
Block 2: 20
Block 3: 50
Block 4: 29
Block 5: 48
Block 6: 28
Block 7: 49

Enter size of each process:
Process 1: 21
Process 2: 34
Process 3: 29
Process 4: 10
Process 5: 53
Process 6: 28
Process 7: 20

Process No.     Process Size    Block No.
1               21              6
2               34              5
3               29              4
4               10              5
5               53              Not Allocated
6               28              7
7               20              2
```